

AI, Human Cognition and Knowledge Collapse

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The paper and the agent's problem

One decision needs two predictions

- ▶ X : precision about the **common** state
- ▶ Y : precision about this **particular** case

Their joint usefulness makes them complements.

Knowledge collapse

Less effort weakens public learning; inherited X_t can converge to zero.

Choose $e \geq 0$; take X, τ_A as given

$$\max_{e \geq 0} f(0, 0) + G(X)\Delta_G + G(X)G(Y)\Delta_X - c(e)$$

$G(X)\Delta_G$: probability $\times$ stand-alone gain

$G(X)G(Y)\Delta_X$: joint probability $\times$ complementarity gain

$c(e)$: resource cost of the chosen effort

$$Y = \underbrace{\sigma^{-2}}_{\text{prior}} + \underbrace{\lambda_I e}_{\text{human signal}} + \underbrace{\tau_A}_{\text{AI signal}}$$

Independent Gaussian evidence $\Rightarrow$ precisions add.

Externality. Effort also creates public precision $\lambda_G E_t$, but a short-lived atomistic agent does not value it.

Main result: complements and substitutes

Economic logic

General knowledge makes case learning more valuable:

$$X \uparrow \Rightarrow MB_e \uparrow.$$

Effort and AI improve the same Y ;
diminishing returns make them substitutes:

$$\tau_A \uparrow \Rightarrow MB_e \downarrow.$$

$$U_e = \lambda_I \Delta_X G(X) g(Y) - e^{1/\varepsilon} = 0$$

$$\underbrace{U_{eX}}_{\text{complementarity}} = \lambda_I \Delta_X g(X) g(Y) > 0$$

$$\underbrace{U_{e\tau_A}}_{\text{substitution}} = \lambda_I \Delta_X G(X) g'(Y) < 0$$

Strict-sign domain

$X, Y, \Delta_X, \lambda_I, \varepsilon > 0$, with $g > 0$ and $g' < 0$.

Boundary. At $X = 0$, $U_{e\tau_A} = 0$ and $g(0)$ is not finite. Strict signs are interior; discrete comparisons preserve the weak global logic.

What I did: test the boundary assumption

1. Audited Observation 1

The mechanism is right; strict cross-partials require $X > 0$.

2. Relaxed $\Delta_I = 0$

Baseline: case-specific correctness has *no* value if the common prediction is wrong.

Useful benchmark, but strong: case knowledge may still improve triage when general knowledge is poor.

Lean audit: derivatives, signs, boundary results, and the conditional extension implication compile in Lean/Mathlib with no proof holes.

Section 5 changes aggregation, synthetic data, or public learning e^β ; it does not relax $\Delta_I = 0$.

Own extension: $\Delta_I > 0$

$$U_e = \lambda_I g(Y) \underbrace{[\Delta_I + G(X)\Delta_X]}_{\substack{\text{stand-alone return} \\ + \text{weighted complementarity}}} - e^{1/\varepsilon}.$$

At $X = 0$, the Δ_I return survives:

$$\Delta_I > 0 \Rightarrow e_\Delta(0, \tau_A) > 0 \Rightarrow F_\Delta(0) > 0.$$

What this proves

Zero is not a fixed point. Positive low-knowledge states and welfare remain open.

Where I did not believe the AI

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Enlarged boundary calculation from the same photograph

Checked claim. Strict complementarity and substitution were read as global derivative statements. At $X = 0$, however, $G(0) = 0$, $U_{eTA} = 0$, and $g(X) \rightarrow +\infty$ as $X \downarrow 0$.

Verdict. Correct on the interior; the strict cross-partial wording needs a boundary-domain qualification.

Full two-page check: interior above,

boundary below